

INFERRING THE DETAILED HISTORY OF A GEOTHERMAL FIELD USING DATA-DRIVEN TECHNIQUES



Source: <https://www.nzgeothermal.org.nz/>

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What is the problem?

Geothermal energy is an important renewable energy source used in Aotearoa and around the world. To protect our geothermal reservoirs, it is important to conduct reservoir modeling, to ensure operators do not over-extract from a geothermal resource, and thus damaging it. However, there is often not enough data provided for modeling to be effective.

Modelers need to know the amount of mass extracted from a certain well across time, however this is rarely known. Occasionally, TFT measurements are taken, which can provide this data, but for remaining timepoints, modelers are forced to guess what the extraction was, leading to less accurate, unreliable predictions. The purpose of this project is to develop a model which can take in the known TFT measurements and known bulk extraction values (which are measured at every timepoint at a separator), and output a detailed history for each individual well. Fig 1 shows how bulk values at a separator are often known, but values for individual wells aren't.

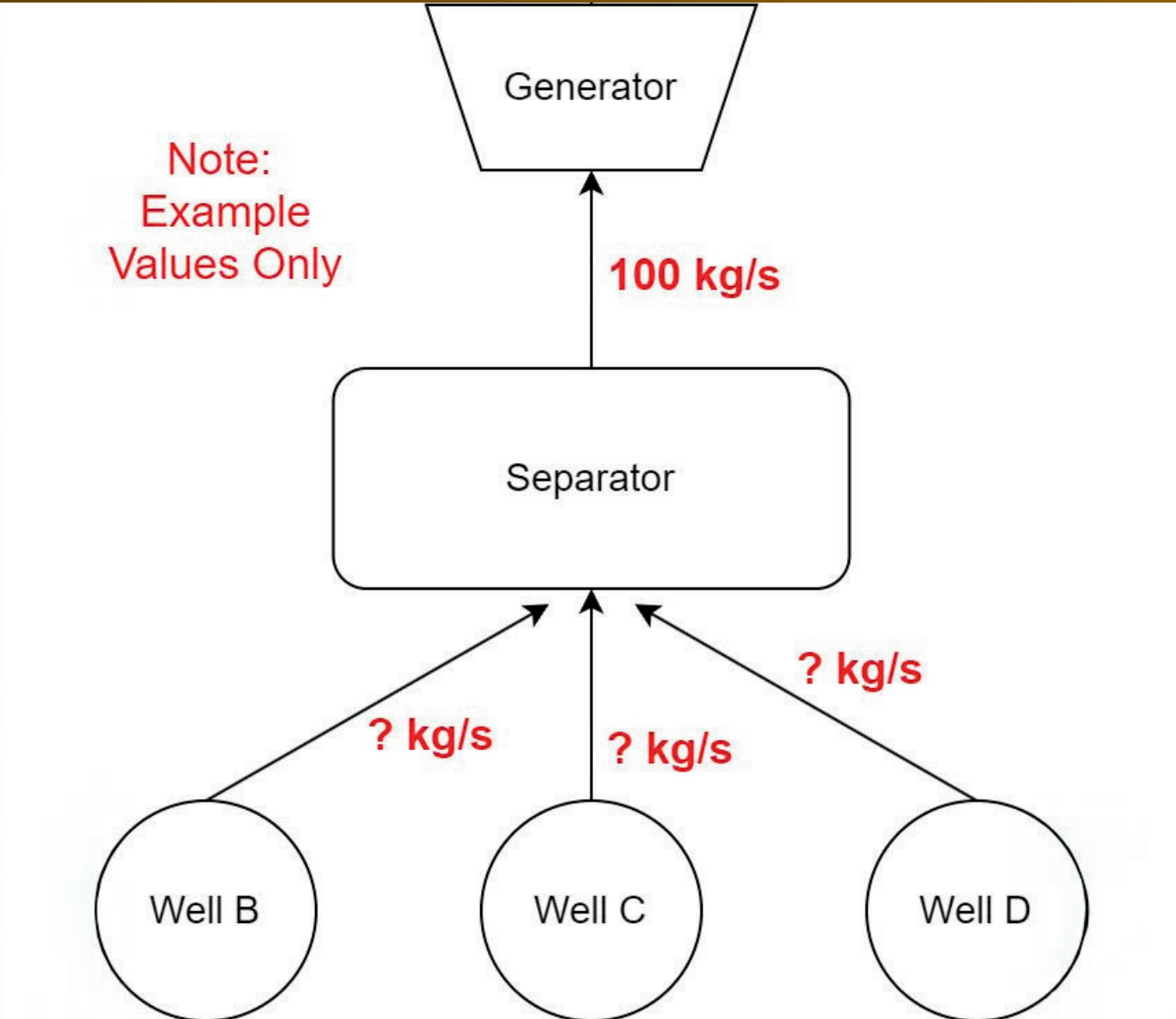


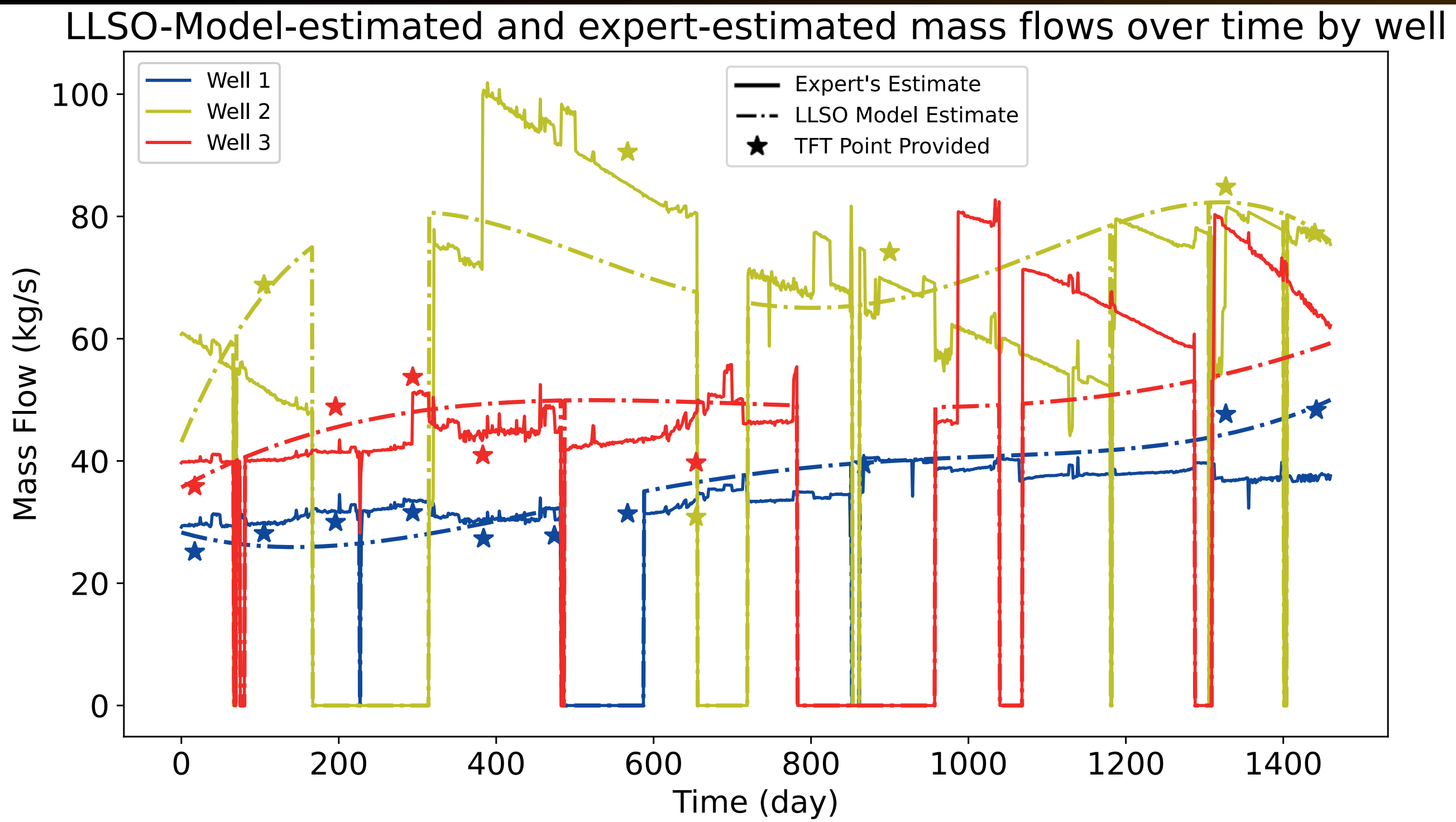
Figure 1: Example of geothermal field mass flow rates

Linear Least Squares Optimisation (LLSO)

By treating the situation as an inverse problem, it is possible to reformulate this into an optimisation problem. In linear least square optimisation, the parameters being solved for are chosen to be coefficients of a polynomial model that represents the mass flow over time for each well. This linear problem can be solved in a least-squared fashion to give an optimal polynomial equation for each well.

Iterative Model Improvement Process:

Initially, this was designed for constant models, then linear models, and then expanded to allow the polynomial to be of an arbitrarily high degree. For higher degree polynomials, the model began **overfitting** on the provided TFT points. By including first-derivative Tikhonov regularisation, we can ensure that the model does not overfit to the provided data by 'smoothing' out the results. A Particle Swarm Optimisation metaheuristic was used to determine the optimal parameter values which give the best results. Output from the LLSO model on a **real** geothermal dataset is shown below:



Discussion and Future Work:

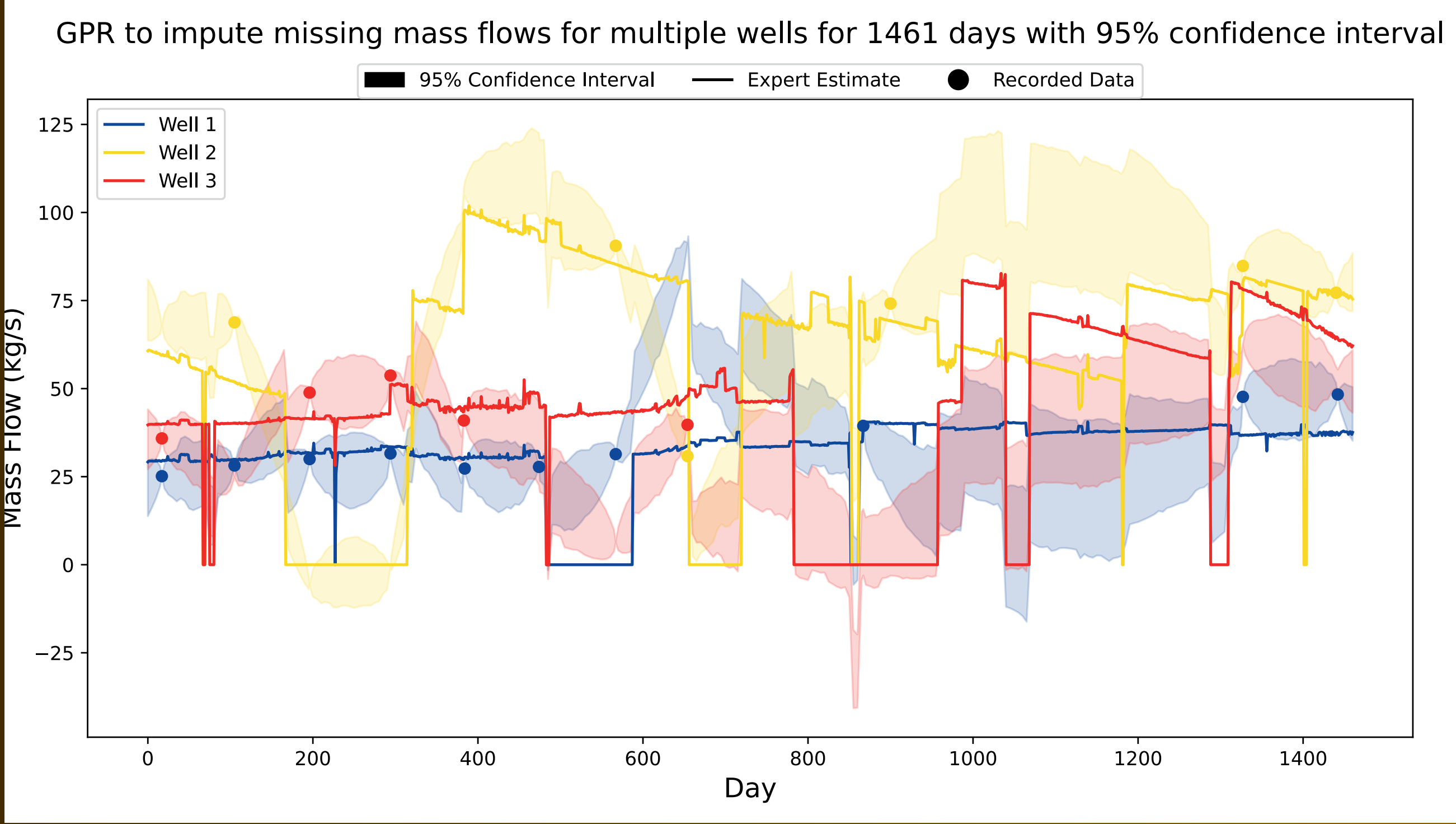
LLSO gives a feasible, reliable estimate of a geothermal field's extraction history. When using a simulated dataset, it modeled the true data with high accuracy. For real data, there is no true data to compare to and provided data is often more sparsely provided, but LLSO still provides an estimate which mostly aligns with that of a geothermal expert. To improve this model, future work could involve enforcing non-negativity, investigating other forms of regularisation, including periodic activity using trigonometric functions, and having weightings based on the confidence of TFT points.

Gaussian Process Regression (GPR)

Gaussian process regression (GPR) is a method used for interpolating missing values using correlations between known data points. These correlations drive the width of the uncertainty intervals produced where higher correlations create narrower intervals, and vice versa. GPR needs a prior mean and a kernel, commonly known as a covariance matrix, to be specified. The kernel defines how points are correlated with each other.

Iterative Model Improvement Process:

Initially the problem solved was very simple with there existing only two wells in the geothermal field and those two wells always being on. Assumptions were then relaxed along the way as more wells were added in and wells could now switch on and off. The latest iteration is the most complex version where real-life data is used to represent a realistic expectation of performance on problems geothermal operators are likely to face. Output from this GPR model is shown below:



Discussion and Future Work:

This research has laid down solid foundations for using GPR to solve sparse data problems in a geothermal context. By enforcing a non-negative constraint on the models, erroneous predictions will be reduced. Creating a custom kernel to best suit specific data would allow for suitable correlations between all data points, likely reducing unwanted oscillations.